🐄 Dairy: Reading · ANSWER KEY

Ag Science 1 · Unit 6 · Level 3 (closer to the class work) · goes with Dairy Production + Dairy Products notes, Darigold video, Cheese presentation, Greek Yogurt, Steps of Ice Cream

Name: _____

From Cow to Carton

Milk and dairy products are Washington's number two farm product, after apples; milk was worth about $1.2 billion in 2017. The U.S. has about 9.3 million dairy cows. In 50 years the number of dairy cows dropped more than 50%, but milk per cow rose more than 130%.

The Holstein, black and white, makes up about 80% of registered U.S. dairy cattle and gives the most milk. The small brown Jersey gives milk high in fat. Other breeds are Guernsey, Brown Swiss, Ayrshire, Red and White and Milking Shorthorn. A calf's first milk, colostrum, has antibodies that protect it, so calves need it soon after birth.

Milk is about 87% water, plus fat, protein (casein), lactose (milk sugar) and minerals. It is a key source of protein, calcium and riboflavin. When a cow's teat is touched, the brain releases oxytocin, which lets milk down. Pain, loud noise or stress can block let-down, so calm cows give more milk. Cows are milked by machine in a milking parlor.

The milk truck driver checks that milk is under 40°F, checks the smell and takes a sample to test for bacteria, fat and antibiotics. At the plant milk is pasteurized, heated to kill harmful bacteria (most often 161°F for 15 seconds), and homogenized, which breaks up fat so cream does not rise. Vitamin D is added. Pasteurized milk still must stay below 40°F.

Milk becomes many products. Cheese forms when casein clumps (coagulates). Yogurt contains probiotics; Greek yogurt is strained to be thicker. Butter is made from cream. Ice cream must have at least 10% milk fat, and the air whipped in is called overrun.

Check your reading

1. Why are there fewer cows but more milk than 50 years ago?
Each cow gives much more milk (better breeding, feed and care).

2. What is the difference between pasteurizing and homogenizing?
Pasteurizing heats milk to kill harmful bacteria; homogenizing breaks up fat.

3. Why do calves need colostrum?
It has antibodies that protect the newborn calf.